Journal of Machine Learning Research 27 (2026) 1-13 Submitted 1/26; Revised 6/26; Published 9/26

Coreference Resolution without Global Attention via Quotient Graph Inference

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Editor: TBD

Abstract

Modern approaches implicitly assume that global coreference resolution requires quadratic attention over the entire text sequence. This paper explores the unwritten assumption and examines whether global token-level cross-attention can be circumvented when features are factorized into distinct, architecturally separated representation channels. Our results indicate that quadratic token interaction can be restricted to localized context windows without severe performance degradation when combined with factorized feature channels and connections are re-established between clusters among different windows. Long-range entity structure is subsequently recovered through a Graph Neural Network running over a compressed quotient graph of locally induced equivalence classes, running at true quadratic cost $\mathcal{O}(C^2)$ over $C \ll M$ (number of mentions) clusters. Our core contribution is this structural decomposition principle that allows us to avoid training one model to learn all features. In our implementation, we use a couple of universal models as a realistic source of these distinct feature channels, while all inference is performed by lightweight learned components containing 16.1M parameters. Evaluated on the CoNLL-2012 benchmark with gold mentions, our approach achieves a CoNLL F_1 score of 0.8630 on the test split (0.8668 on validation). This indicates that strong document-level coreference performance can be achieved without global text-level attention through distinct channel factorization and structural quotient-graph inference.

Keywords: coreference resolution, representation factorization, quotient graph, Graph Neural Networks, window-based inference, transitive closure, efficient inference

1 Introduction

Coreference resolution is defined as the problem of clustering a set of entity mentions $\mathcal{M} = \{m_1, \ldots, m_n\}$ into groups that are represented by one *canonical entity*. Formally, this induces a global equivalence relation $\sim$ over these mentions, where $m_i \sim m_j$ if and only if they refer to the same canonical entity. The true target output of the system is a partition $\mathcal{E} = \mathcal{M}/\sim$. From this perspective, coreference can be viewed as a structured prediction problem whose objective is to recover a latent quotient structure rather than to independently classify mention pairs or model token-level interactions.

In this work, we explicitly adopt this equivalence-relation formulation as the primary object of inference and study how it can be recovered from localized computations over restricted subsets of $\mathcal{M}$. We propose viewing the problem as a two-level decomposition: (i) construction of partial equivalence relations under localized observation constraints, and (ii) recovery of the global partition via composition and closure over these local relations in a compressed representation space. This formulation separates local relational induction from global consistency enforcement, allowing long-range dependencies to be recovered through structured aggregation over an intermediate quotient graph of locally induced equivalence classes rather than direct global interaction over the full mention set.

Crucially, this work departs from the paradigm that long-range entity coreference requires monolithic, end-to-end token contextualization. Our results suggest that a substantial portion of the apparent reliance on global sequence-wide attention may arise from representation entanglement—where semantic, contextual, and discourse features are melted into a single joint vector space. Our results suggest that factorizing the required global signal into distinct representation channels may permit the isolation of quadratic sequence operations to tight local windows, shifting the burden of long-range consistency from token-level attention to structural inference over a compressed quotient graph of locally induced equivalence classes.

The key diagnostic that launched this work is an empirical probe of the representation space: taking the two axis vectors for pairs of different-lemma mentions, their raw cosine similarity separates coreferent from non-coreferent pairs at chance (AUC ≈ 0.57). The *same* vectors, read by a small learned linear probe, separate them at AUC $= 0.91$. This signal is fully present in the uncollapsed axes; reducing either to a scalar similarity measure discards discriminative structure.

The target decision boundary explicitly relies on the joint interpretation of these un-fused coordinate streams. Furthermore, we demonstrate that external symbolic channels—such as discrete lexical match dimensions—must not be collapsed early into scalar additions or post-hoc decoding heuristics. When integrated nonlinearly via vector concatenation directly into the relational pairing heads, these side-channels unlock significant downstream coordinate synergy. By replacing independent pairwise matching heuristics with an explicit Graph Neural Network (GNN) operating over the quotient space, global consistency and competitive performance are recovered at an absolute computational complexity class completely decoupled from the document's total mention count.

The remainder of this paper is organized as follows. Section 2 reviews related work and contextualizes our contribution. Section 3 outlines the overall workflow of the two-stage pipeline. Section 4 develops the mathematical foundations of representation factorization. Sections 5 and 6 detail the technical implementation of the Stage A local window inference procedure and Stage B quotient-space matching modules respectively. Section ?? describes experimental setup, and Section 7 presents empirical results and analysis.

2 Related Work

Scaling coreference resolution to long documents has traditionally relied on mitigating the quadratic cost of sequence-wide self-attention using two primary techniques:

Modern architectures frequently bypass full self-attention by substituting localized or sparse attention matrices, such as sliding windows and block-diagonal grids (Caciularu et al., 2023). While these methods reduce memory consumption during representation extraction, they remain structurally bound to an end-to-end fine-tuning paradigm where entangled relational features must be propagated across deep sequence lengths. In contrast, our approach restricts quadratic calculations strictly to isolated local execution windows, offloading long-range tracking to an uncontextualized cluster graph layer.

To bypass large all-pairs mention scoring matrices, several pipelines implement coarse-to-fine filtering heuristics (Dobrovolskii, 2021; Bohnet et al., 2023) or recursive mention clustering hierarchies (Luo et al., 2025). While effective at pruning computational overhead, these systems are vulnerable to cascading errors during iterative merging steps. We depart from this by reformulating cross-window resolution as a formal completion task over a highly compressed quotient graph. By leveraging an un-fused coordinate basis rather than early scalar similarities, our method preserves fine-grained channel identities over extreme token distances without requiring global sequence contextualization.

3 Solution Overview

To bypass global quadratic text attention, the framework processes an incoming document through a two-stage hierarchical inference pipeline that operates first on bounded local context and then on a compressed quotient representation of induced entities.

1. **Decoupled Encoding Stream:** Each mention is mapped into two fixed, pretrained representation spaces capturing distinct semantic and contextual information. These encoders are frozen throughout training, and their outputs are never fused into a single joint representation at the token level. All subsequent reasoning is performed over these fixed, separated feature views.

2. **Stage A (Intra-Window Relational Construction):** The document is partitioned into a set of non-overlapping windows of fixed size. Within each window, all pairwise mention interactions are computed using a local scoring function over the distinct representations. A transitive inference procedure

(union-find) is then applied strictly within each window, producing a set of locally consistent equivalence structures.

3. **Quotient Representation Induction:** The output of Stage A is a collection of locally inferred equivalence groups, each representing a collapsed subset of mentions. These groups are treated as atomic nodes for all subsequent computation, inducing a reduced representation space in which reasoning operates over clusters rather than raw mentions.

4. **Stage B (Cross-Window Quotient Matching via GNN):** Global entity consistency is recovered by evaluating structural interactions over the induced clusters in the form of an antecedent ranking task. A Graph Neural Network performs message passing over the quotient graph topology, processing size-invariant node features derived from local non-parametric cluster aggregations combined with concatenated lexical signals. Final document-level entities are obtained by computing the transitive closure over this pointer graph, yielding a globally consistent partition of mentions.

4 Representation Factorization Framework

Monolithic language models rely on end-to-end contextualization over full-document sequences to implicitly disentangle semantic, discourse, and referential dependencies. This couples representation learning with relational inference and induces unnecessary computational cost. We instead hypothesize that coreference can be reformulated as a structured inference problem over localized observations combined through a compressed global reasoning space. Under this view, global entity structure does not require direct token-level interaction, but can be recovered from compositions of locally induced relational evidence. This motivates a factorization of inference into distinct representation channels that preserve complementary signals for downstream relational reconstruction.

Assumption 1 (Distinguishable Feature Representation Basis) *Let $\mathcal{V}$ denote the complete decision space required to resolve entity identity. We assume that discriminative identity information can be systematically recovered across d distinct, architecturally separated feature channels whose signals remain structurally intact when preserved separately:*

$$\mathcal{V} = \mathcal{V}_{context} \times \mathcal{V}_{semantic} \times \dots$$

where individual channels preserve specialized signals that are obscured when compressed into low-dimensional similarity measures or early aggregation operations.

$$\mathbf{h}_i = \left[P_{\text{ctx}}\left(f_{\text{RoBERTa}}(m_i)\right) \parallel P_{\text{bge}}\left(f_{\text{BGE}}(m_i)\right)\right] \in \mathbb{R}^{d_1+d_2}$$

where $P_{\text{ctx}} : \mathbb{R}^{2048} \to \mathbb{R}^{1024}$ and $P_{\text{bge}} : \mathbb{R}^{1024} \to \mathbb{R}^{1024}$ are learned linear projections mapping frozen activations into isolated coordinate paths.

To isolate whether this specific factorization drives system capability, we introduce a channel control configuration channel $\in \{\text{both}, \text{ctx}, \text{bge}\}$. When a single channel is isolated, the unused projection layer is dropped entirely and the upstream scoring heads are dynamically resized. This architecture guarantees that single-channel variants are trained completely from scratch rather than evaluating zeroed vectors at inference time, avoiding confounding the capacity of the model head with the signal density of the channel.

The structural necessity of this factorization is verified through rigorous single-channel control experiments. Isolation of the contextual channel (channel = ctx) drops full global performance to 0.8370 F_1, while isolating the semantic channel (channel = bge) triggers an absolute collapse to 0.7677 F_1. This confirms that the target decision boundary explicitly relies on the joint interpretation of un-fused coordinate streams. Pruning either single coordinate path prevents the model from mapping cross-window dependencies effectively.

The critical property of this system is not that the underlying encoders are independent, but that the decision information is supplied through separate coordinate systems whose signals remain explicitly distinguishable throughout inference. Frozen universal encoders provide one convenient realization of this condition; any mechanism capable of producing comparably distinct channels would satisfy the same architectural requirement. The contribution of this work is therefore not the use of frozen models itself, but the demonstration that preserving factorized information channels can mitigate the reliance on global token-level attention within document-level coreference resolution.

This factorization is explicitly validated by an empirical probe: computing raw cosine similarities over these embedding dimensions directly yields an AUC of ≈ 0.57 on separating coreferent from non-coreferent pairs. Conversely, a shallow learned linear probe trained over the un-fused coordinate vectors $(\mathbf{h}_i, \mathbf{h}_j)$ separates identity at an AUC of 0.91. This demonstrates that substantial coreference information is present in the preserved multi-channel representation and remains linearly recoverable. The large gap between cosine similarity and linear-probe performance suggests that scalar similarity measures discard discriminative structure that remains available when the separate coordinate channels are preserved.

5 Stage A: Bounded Local Equivalence Reconstruction

Given the factorized representations $\mathbf{h}_i$, we restrict initial quadratic interaction to bounded observation domains to evaluate the structural locality of coreference decisions.

Assumption 2 (Transitive Link Reconstruction) *Global entity paths are locally dense: any two coreferent mentions $m_a, m_b \in \mathcal{M}$ are connected via a finite* *chain of intermediate local links, where each adjacent link pair falls within a bounded sequence window.*

The architectural viability of restricting token-level quadratic attention to isolated local windows is supported by the factorization established in Section 4. Because the coordinate representations retain un-collapsed channel signals, local relational boundaries preserve substantial identity information that can drop under conventional scalarized similarity spaces, mitigating the contextual degradation typically observed in naive windowed models. This allows us to apply a deterministic partition over the document into disjoint sequence windows W_k of length $K = 256$ tokens, such that each mention is assigned to exactly one window via a mapping function $\pi : \mathcal{M} \to \mathcal{W}$.

For every pair of mentions co-located within W_k, an antecedent scoring head (`AntecedentScorer`) calculates local compatibility alongside bucketed distance tokens:

$$s(i, j) = \mathrm{MLP}\left([\mathbf{h}_i \parallel \mathbf{h}_j \parallel \mathbf{e}_{\mathrm{dist}(i-j)}]\right)$$

A local equivalence relation $\sim_k$ is induced via thresholding against a learned null bias parameter τ. Computing the transitive closure via a union-find algorithm strictly within each window boundaries yields a set of local equivalence classes:

$$\mathcal{C}_k = W_k / \sim_k^*$$

The union of these sub-clusters across all windows forms our complete intermediate entity space $\mathcal{C} = \bigcup_{k=1}^{W} \mathcal{C}_k$. It is important to note that elements of $\mathcal{C}_k$ constitute local connected components under localized constraints, rather than a final global quotient space. Each local cluster $C \in \mathcal{C}$ is subsequently treated as an atomic node in a quotient graph of locally induced equivalence classes for cross-window inference layers.

Evaluating the isolated performance of Stage A against full-document gold targets (leaving each local cluster as its own unmerged entity) yields a baseline score of 0.7439 F_1 on CoNLL-2012. Crucially, local equivalence reconstruction alone retains the vast majority of final performance without requiring global sequence-wide matching dependencies, trailing the complete system by 0.1191 points. This provides a central empirical finding of this work: a large fraction of document-level coreference structure remains highly concentrated within bounded local neighborhoods.

6 Stage B: Quotient-Space Completion via Graph Neural Networks

The intermediate local equivalence classes derived in Section 5 act as atomic nodes in a highly compressed graph. Rather than executing unconstrained mention-pair matching across window splits, Stage B is formulated as an antecedent ranking problem over the quotient graph topology. The objective is to recover global coreference consistency by establishing a directed cross-window pointer forest over macro-nodes, bounded by a strict complexity class of $\mathcal{O}(M + C^2)$ where $C \ll M$.

Proposition 1 *Let $\mathcal{C} = \bigcup_k \mathcal{C}_k$ denote the set of locally induced equivalence classes where $\mathcal{C}_k = W_k / \sim_k^*$. If an operator $\approx$ correctly identifies cross-window equivalences between elements of $\mathcal{C}$, then the global entity partition $\mathcal{E} = \mathcal{M}/\sim$ is recovered exactly by the transitive closure of $\approx$ lifted back onto $\mathcal{M}$.*

6.1 Quotient Graph Formulation & Node Pooling

Let $\mathcal{C} = \{C_1, C_2, \dots, C_C\}$ be the set of clusters extracted by Stage A across all sequence windows. To map these clusters into the macro-graph without introducing information loss or smearing discriminative features, we implement a size-normalized log-mean-exp pooling operator (`lse`) over the per-mention channel vectors (each frozen stream projected to 512 and concatenated, $\mathbb{R}^{1024}$) to construct base node representations $\mathbf{v}_c \in \mathbb{R}^{1024}$; these are linearly mapped to a 512-dimensional node space and contextualized by the encoder below, and the pairwise head consumes the contextualized $\mathbb{R}^{512}$ node vectors (so $\mathbf{x}_{\text{pair}} \in \mathbb{R}^{2051}$):

$$\mathbf{v}_c = \log\left(\frac{1}{|C_c|}\sum_{m_i \in C_c} \exp(\mathbf{h}_i)\right)$$

This soft-maximum operator preserves the vector identity of the single most discriminative mention within the sub-cluster (e.g., a distinct proper noun surface string), bypassing the signature attenuation or "smearing" penalty typical of uniform attention pooling mechanisms.

6.2 Structural Message Passing & Outer Residuals

The pooled cluster nodes are contextualized globally using a 2-layer Transformer encoder serving as a fully connected quotient message-passing network. To prevent *all-null collapse*—where deep transformer layers over-smooth representations to near-identical vectors (mean pairwise cosine $\to$ 1.000) and completely erase the discriminative features required to resolve distinct entity identities—we enforce a rigid outer residual connection:

$$\mathbf{H} = \mathbf{H}_0 + \text{GNN}(\mathbf{H}_0)$$

where $\mathbf{H}_0$ represents the initial array of un-smoothed cluster node vectors augmented with window position embeddings. The residual path ensures that localized discriminative structures always reach the final scoring heads, shifting the role of graph message passing from base representation creation to the relational refinement of existing entity boundaries.

6.3 Concatenated Lexical Integration

To assist the neural coordinates across vast token spans, symbolic side-channels are integrated directly into the pair interaction MLP. Crucially, experimental validation reveals that collapsing discrete lexical matching features into an additive scalar penalty or a post-hoc decoding heuristic creates threshold coupling that degrades performance on non-lexical categories. We instead concatenate the symbolic signals directly into the joint neural feature space:

$$\mathbf{x}_{\text{pair}} = [\mathbf{v}_A \parallel \mathbf{v}_B \parallel \mathbf{v}_A \odot \mathbf{v}_B \parallel |\mathbf{v}_A - \mathbf{v}_B| \parallel \mathbf{x}_{\text{lex}}]$$

where $\mathbf{x}_{\text{lex}} \in \mathbb{R}^3$ tracks:

1. **IDF-weighted token Jaccard similarity:** $\sum_{t \in A \cap B} \text{idf}(t) / \sum_{t \in A \cup B} \text{idf}(t)$, where document-local $\text{idf}(t) = \log(M/(1 + \text{df}(t))) + 1$.
2. **Substring containment:** A binary indicator marking if any content mention (excluding bare pronouns) in A is a substring of any content mention in B, or vice versa.
3. **Exact mention match:** A binary flag tracking if the two clusters share an identical content mention surface text string.

This concatenation allows the relational head to interpret symbolic signals non-linearly and conditionally based on the surrounding neural context, unlocking a +0.62 F_1 improvement over non-concatenated variants. Final entity chains are recovered by allowing each macro-node to predict a single antecedent pointer among strictly earlier temporal windows, thresholded against a learned null node τ.

7 Results and Analysis

All evaluation experiments are conducted on the test split of the CoNLL-2012 English coreference benchmark (Pradhan et al., 2012) utilizing gold mention boundaries. Following empirical data mixture optimizations, our lightweight scoring modules are trained on a uniform corpus mixture consisting of CoNLL-2012, LitBank, CorefUD, and a subsampled partition of 8,000 documents from PreCo. Training enforces a uniform loss across all datasets without negative window-pair subsampling to preserve distributional balance. All baseline results cited from prior literature correspond strictly to their respective gold-mention evaluation configurations.

We isolated the performance of individual channels against the full dual-channel baseline. All variations utilize an identical window configuration ($K = 256$) and the Cluster-GNN architecture trained on the uniform multi-corpus mixture. Table 1 details the results.

The empirical trajectory reveals a clear asymmetry between the two coordinate views. The contextual channel (`ctx`) is heavily dominant, achieving 0.8370 F_1 on

Channel Configuration	Stage A	Stage A+B
Both Channels (BGE + RoBERTa)	**0.8970**	**0.8630**
RoBERTa Contextual Only (`ctx`)	0.8781	0.8370
BGE Semantic Only (`bge`)	0.7960	0.7677

Table 1: Channel ablation control results (CoNLL F_1) on CoNLL-2012 test data splits.

its own—within 3.0 points of the complete system. Conversely, the semantic channel (`bge`) drops significantly to 0.7677, showing that cross-window cluster bridging retains very little tracking signal when divorced from contextual sequence history.

Granular cluster analysis uncovers a strict architectural division of labor between the channels. Pronominal groups (`PRON-only`) are almost completely dependent on the contextual channel, where the `ctx`-only run matches the full architecture baseline (0.6556 vs. 0.6640), while the semantic `bge`-only run collapses to 0.5173. Conversely, proper nouns (`PROPN-only`) lean securely on semantic assignments, where `bge`-only maintains a stable footing (0.7276) relative to context alone (0.7983). This split confirms that pronouns ride the contextual wave, while surface variations of proper nouns rely on global text-invariant semantic spaces.

7.1 Benchmarks

Table 2 establishes the primary system performance evaluated against state-of-the-art architectures using identical gold mention contexts.

Approach	CoNLL	MUC	B^3	CEAFe
Stage A Only (No Merge Baseline)	0.7439	0.8460	0.6910	0.6950
Stage A + B GNN ($\mathcal{O}(C^2)$ + Concatenated Lexical)	**0.8630**	**0.9240**	**0.8409**	**0.8241**
Dobrovolskii (2021)	0.8180	0.8570	0.8000	0.7970
Caciularu et al. (2023)	0.8360	0.8740	0.8200	0.8130
Bohnet et al. (2023)	0.8380	0.8760	0.8220	0.8150
Luo et al. (2025)	0.8510	0.8890	0.8350	0.8290

Table 2: Full-document CoNLL-2012 test results on **gold mention boundaries**. Baseline systems are mapped strictly to their gold-mention settings to isolate linking capability.

Cluster Category Type	CoNLL	MUC	B^3	CEAFe
PROPN-only	0.8377	0.8367	0.8352	0.8392
PRON+NOUN	0.7734	0.7658	0.7526	0.8017
NOUN-only	0.7710	0.7756	0.7703	0.7671
PRON+PROPN	0.7233	0.7390	0.6852	0.7457
NOUN+PROPN	0.7058	0.7104	0.6972	0.7100
all-mixed	0.6996	0.7578	0.6600	0.6809
PRON-only	0.6792	0.7085	0.6543	0.6748

Table 3: Definitive per-cluster category breakdown for the Stage A + B GNN pipeline, CoNLL-2012 test.

7.2 Complexity and Operational Scale

Since window size is fixed, total sequence encoding complexity runs at $\mathcal{O}(WK^2) = \mathcal{O}(NK)$, avoiding full-sequence token quadratics. Within Stage A, localized mention scoring operates at $\mathcal{O}(W \cdot (M/W)^2) = \mathcal{O}(M^2K/N)$.

For Stage B cross-window matching, the integration of macro-node pooling maps the trainable computation path directly into the number of locally induced sub-clusters:

$$T_{\text{Stage B}}(C) = \mathcal{O}(M) + \mathcal{O}(C^2)$$

where C represents the total number of clusters generated by Stage A per document. Because $C \ll M \ll N$, the absolute computation budget required for global entity synthesis is entirely decoupled from the quadratic scale of the document sequence or token length.

Space Layer	Graph Nodes	Evaluation Points	Algorithmic Mode
Token Sequence Space	617.4	381,183	$\mathcal{O}(NK)$ Windowed Attention
Mention Graph Space	57.1	3,260	Unwindowed Mention Baseline
Quotient Graph Space	**20.7**	**428**	$\mathcal{O}(C^2)$ **Trainable Graph Decisions**

Table 4: Average document graph dimensions and operational scales evaluated on CoNLL-2012 test data splits.

Empirical metrics collected on the CoNLL-2012 test set (Table 4) validate this structural optimization. Mean scales establish document dimensions of $N = 617.4$

tokens, $M = 57.1$ nominal mentions, and $C = 20.7$ locally induced macro-nodes. Stage A contracts the entity node space down to 36.3% of its uncollapsed volume, yielding a 2.76:1 node compression ratio.

This contraction drastically reduces the learned evaluation space. While an unwindowed all-pairs system must parameterize and score 3,260 potential mention edges, Stage B operates over a highly compressed topology of only 428 candidate macro-node interactions. This represents an empirical **7.6× structural reduction** in the learned decision graph size. By handling the internal `lse` member pooling as a non-parametric aggregation over cached coordinate segments, the parameter execution path is restricted to a compact macro-node topology.

Our experimental findings indicate that the explicit core theoretical assumptions are empirically viable:

1. **Preservation of Feature Distinctness (Assumption 1):** Reaching an F_1 score of 0.8630 is consistent with the premise that preserving distinguishable feature channels retains structural identity information that remains accessible for downstream reconstruction. This observation is supported by the notable performance gap between raw cosine evaluation and the un-fused multi-channel linear probe.

2. **Validation of Locality-Transitivity (Assumption 2 & Proposition 1):** Stage B achieves an Edge Precision of 78.1% and Edge Recall of 68.6% under the pooled head baseline, which scales up to highly competitive margins when processed via the quotient graph network. Combined with an average fragmentation profile of only 1.45 nodes per true entity, this tracks the accuracy of macro-transitive lifting across window boundaries.

By locking the primary encoder backbones and routing cross-window connections through the compressed quotient graph, the model significantly reduces computational overhead. As itemized in Table 5, trained parameters account for only 2.28% (16.1M) of the total 706.13M parameter budget.

Component	**Parameter Count**
Stage A (AntecedentScorer)	8.43M
Stage B (ClusterGNN)	7.70M
RoBERTa-large	355.00M
BGE-large	335.00M
Total Parameters	**706.13M**

Table 5: System parameter budget distribution.

We note an explicit architectural distinction regarding our computational profile: the primary claim of this work is not that the overall pipeline contains zero global-attention calculations, as the frozen backbones utilize internal transformer layers.

Architecture Model	Inference Speed	Incremental Allocated	Total Reserved VRA
Quotient-GNN ($\mathcal{O}(C^2)$)	19.4 ms / 1k tok	64.2 MB	2.41 G
Unwindowed Baseline ($\mathcal{O}(M^2)$)	**13.8 ms / 1k tok**	198.5 MB	2.54 G

Table 6: Runtime and memory profile measurements collected per-document during inference on a single RTX 4060 Laptop GPU.

Rather, the claim demonstrates that **task-specific entity inference and downstream cluster assembly are executed completely without document-wide token interaction**.

In inference, this architectural factorization limits memory usage. The Quotient-GNN restricts peak active incremental allocations to 64.2 MB by preventing the materialization of the full unwindowed mention-pair matrix, yielding a **3.1**$\times$ optimization in active running footprint over an unwindowed setup. Total cumulative VRAM footprint remains bounded near 2.41 GB due to the shared memory overhead of the frozen token embedding caches (200 MB) and CUDA driver baselines.

An honest evaluation of wall-clock times reveals that at standard lengths (mean 617.4 subtokens), the unwindowed architecture retains a minor speed advantage (13.8 ms vs 19.4 ms per document). Because sequence spans are short in this corpus, unwindowed $\mathcal{O}(N^2)$ mechanics are not yet computationally punitive, allowing an unwindowed layout to benefit from highly parallel vectorized GPU paths. The windowed system encounters a minor Python execution loop penalty when processing small windowed batches sequentially. The asymptotic runtime advantage of the $\mathcal{O}(NK)$ design manifests on significantly longer document horizons, while the memory runtime ceiling reduction remains immediately visible.

8 Conclusion

Our findings demonstrate that global entity resolution can be successfully recovered from local inference paired with quotient-graph reconstruction. Our results suggest that document-level coreference can be performed without global token-level attention when entity inference is reformulated as local equivalence reconstruction followed by quotient-space completion. In our implementation, preserving separate representation channels was sufficient to recover global entity structure through local edge induction and quotient-space reconstruction without requiring document-wide token attention.

These results raise the possibility that some relational prediction problems may admit similar structural decompositions into localized inference and quotient-space reconstruction, particularly where decision boundaries can be expressed as stable compositions of distinguishable information channels. In the setting studied here, exhaustive sequence-wide cross-attention does not appear to be an absolute prerequisite for strong document-level coreference performance due to the inherent equivalence relation structure. Under this decentralized processing paradigm, downstream multi-task architectures could allocate parameter capacity toward isolating distinct specialized feature views, leveraging a lightweight central language infrastructure purely to execute structural orchestrations and macro-equivalence pooling rather than computing dense all-to-all token interactions over an entire contextual window.

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